

Launchpad Astrophysics Summer Program

Astrophysics Foundation Assignment 1

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Using High-School Physics and Mathematics to Explore the Universe

Instructions

- Attempt all problems.
- Show detailed calculations and physical reasoning.
- You may use a scientific calculator.
- Assume SI units unless stated otherwise.

Useful Physical Constants

Quantity	Value
Gravitational constant G	$6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
Speed of light c	$3.0 \times 10^8 \text{ m/s}$
Mass of Sun $M_{\odot}$	$2.0 \times 10^{30} \text{ kg}$
Radius of Sun $R_{\odot}$	$7.0 \times 10^8 \text{ m}$
Luminosity of Sun $L_{\odot}$	$3.8 \times 10^{26} \text{ W}$
Astronomical Unit (AU)	$1.5 \times 10^{11} \text{ m}$
Stefan-Boltzmann constant σ	$5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$
Boltzmann constant k_B	$1.38 \times 10^{-23} \text{ J/K}$

Problem 1: Escape Velocity from a Neutron Star

Neutron stars are among the densest known objects in the universe. Consider a neutron star with:

$$M = 1.4M_{\odot}$$
$$R = 12\text{ km}$$

- (a) Derive the expression for escape velocity from the surface of a spherical body.
- (b) Calculate the escape velocity from the neutron star.
- (c) Express your answer as a fraction of the speed of light.
- (d) Explain physically why Newtonian mechanics may begin to fail for such compact objects.

Hint:

Use conservation of energy:

$$\frac{1}{2}mv^2 = \frac{GMm}{R}$$

Problem 2: Temperature of a Star Using Blackbody Radiation

A star behaves approximately like a blackbody radiator.

Suppose a star has:

$$L = 25L_{\odot}$$
$$R = 3R_{\odot}$$

The luminosity of a star is related to temperature by the Stefan-Boltzmann law:

$$L = 4\pi R^2 \sigma T^4$$

- (a) Calculate the surface temperature of the star.
- (b) Compare your answer with the Sun's surface temperature ($\approx 5800\text{ K}$).
- (c) Explain what happens to the emitted radiation as the temperature increases.

Problem 3: Orbital Motion of an Exoplanet

An exoplanet orbits a Sun-like star at a distance of:

$$r = 0.05\text{ AU}$$

Assume the star has the same mass as the Sun.

- (a) Using Newton's law of gravitation and circular motion, derive the expression for orbital velocity.
- (b) Calculate the orbital speed of the exoplanet.

- (c) Determine the orbital period of the exoplanet.
- (d) Compare the result with Earth's orbital period and explain why some exoplanets have extremely short years.

Hint:

Equate gravitational force and centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

Problem 4: Energy Released by Accretion onto a Black Hole

Matter falling toward a black hole releases enormous amounts of energy.

Suppose a gas cloud of mass:

$$m = 2 \times 10^{12} \text{ kg}$$

falls into an accretion disk around a black hole.

Assume 10% of the rest mass energy is converted into radiation.

- (a) Calculate the total rest mass energy of the gas cloud.
- (b) Determine the energy emitted as radiation.
- (c) Compare this energy with the annual energy consumption of humanity ($\sim 6 \times 10^{20} \text{ J}$).
- (d) Explain why accretion onto compact objects is one of the brightest phenomena in astrophysics.

Hint:

Use Einstein's mass-energy relation:

$$E = mc^2$$

Problem 5: Doppler Shift of Light from a Receding Galaxy

A distant galaxy emits light with wavelength:

$$\lambda_0 = 500 \text{ nm}$$

An observer on Earth measures the wavelength to be:

$$\lambda = 520 \text{ nm}$$

- (a) Calculate the redshift z of the galaxy.
- (b) Determine the recession velocity of the galaxy assuming:

$$z = \frac{v}{c}$$

valid for small velocities.

- (c) Explain physically why distant galaxies appear redshifted.
- (d) Discuss how redshift measurements helped establish the idea of an expanding universe.

Final Exploratory Question (Optional)

Which topic in this assignment fascinated you the most?

Write a short paragraph describing:

- what concept interested you,
- what confused you,
- and one astrophysical question you would like to explore further.

"The universe is under no obligation to make sense to us."

— Neil deGrasse Tyson